

Significant Error: Citations to Research With Publicized Statistical Errors

Ken Cor Gaurav Sood

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Abstract

Publicizing a statistical error could sharply reduce reliance on affected papers or prompt researchers to qualify their citations. We examine both responses after a 2011 critique of a common mistake in neuroscience: inferring that two effects differ because one is statistically significant and the other is not. Papers flagged for the mistake continued to receive citations well beyond a handful of highly cited cases. Their median annual citations rose from 5 in 2010 to 13–17 during 2012–2015; comparison papers rose from 4 to 10–11. In a separate sample of citations to flagged papers, 94 of 95 valid completed ratings recorded no acknowledgment of concerns. These findings document widespread continued use with little recorded qualification. Comparisons with unflagged papers leave the size of any citation penalty uncertain; continued citation does not establish that publicity had no effect.

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1 Introduction

Publicizing a statistical error gives researchers a reason to reconsider their reliance on affected findings. A substantial response could take two forms: fewer citations to papers containing the error, or more citations that acknowledge the problem. We examine both after a prominent methodological critique in neuroscience.

The mistake is to treat a statistically significant effect in one group and a nonsignificant effect in another as evidence that the groups differ. That conclusion requires a test of the difference itself. Significance in one group can reflect greater precision rather than a larger effect (Gelman and Stern, 2006). The missing test leaves the claimed contrast unsupported by that argument; it does not establish that every conclusion in the paper is false.

Nieuwenhuis et al. (2011) identified 157 papers in which a comparison of effects was relevant and classified 79 as making the mistake. They supplied us with the paper-level classifications. Researchers who understand the critique can recognize the mistake in an original paper and reconsider its use. The critique therefore need not name each affected paper to influence citations.

The affected papers remained widely cited, and recorded qualifications were rare. Median as well as mean citations increased after the critique; 69 of 76 flagged papers received more citations per year in 2012–2015 than in 2010. Almost all valid completed ratings in our citation-context sample recorded no concern. The comparison with unflagged papers is less conclusive about how much publicity changed citation growth. We distinguish that causal question from the observed persistence of largely unqualified citation.

2 Data and comparison

The source review covers behavioral, systems, and cognitive neuroscience papers published in 2009–2010 in *Nature*, *Science*, *Nature Neuroscience*, *Neuron*, and sampled issues of the *Journal of Neuroscience*. The classification concerns the particular statistical inference reviewed, not the validity of every result in a paper.

The archived citation exports cover 155 of the 157 classified papers. Two further histories contain the intended cited paper itself and records predating it, indicating invalid citation searches. Excluding these histories leaves 76 flagged and 77 comparison papers. We count distinct citing documents per paper and year, retain citation-free years within covered histories, and exclude the incomplete 2016 export. Appendix A documents coverage, duplicates, and linkage checks.

The critique appeared online on August 26, 2011. We use 2010 as the baseline and 2012–2015 as the post-publicity period, omitting the transition year 2011. The main regression is Poisson pseudo-maximum likelihood with article and calendar-year fixed effects. It compares proportional citation changes in flagged and comparison papers; uncertainty is clustered by article. A relative post/pre citation ratio below one indicates slower proportional growth for flagged papers. Appendix B gives the specification and alternative comparisons.

Interpreting that contrast as an effect of publicity requires comparable proportional citation trajectories without the critique. Error status was not randomized, and papers may differ in topic,

prominence, and ordinary citation growth. A general warning could also affect both groups; the comparison concerns their differential response, not the critique’s total influence on science.

3 Continued citation and qualification

3.1 Citation trajectories

Continued citation extends beyond a few prominent papers. Flagged papers’ median annual citations rose from 5 in 2010 to between 13 and 17 in each year from 2012 through 2015 (Figure 1). Comparison papers’ medians rose from 4 to 10–11. Mean citations also rose: from 9.4 to a post-period annual average of 21.4 for flagged papers, and from 6.5 to 16.7 for comparison papers. For 91% of flagged papers, the post-period annual average exceeded the baseline.

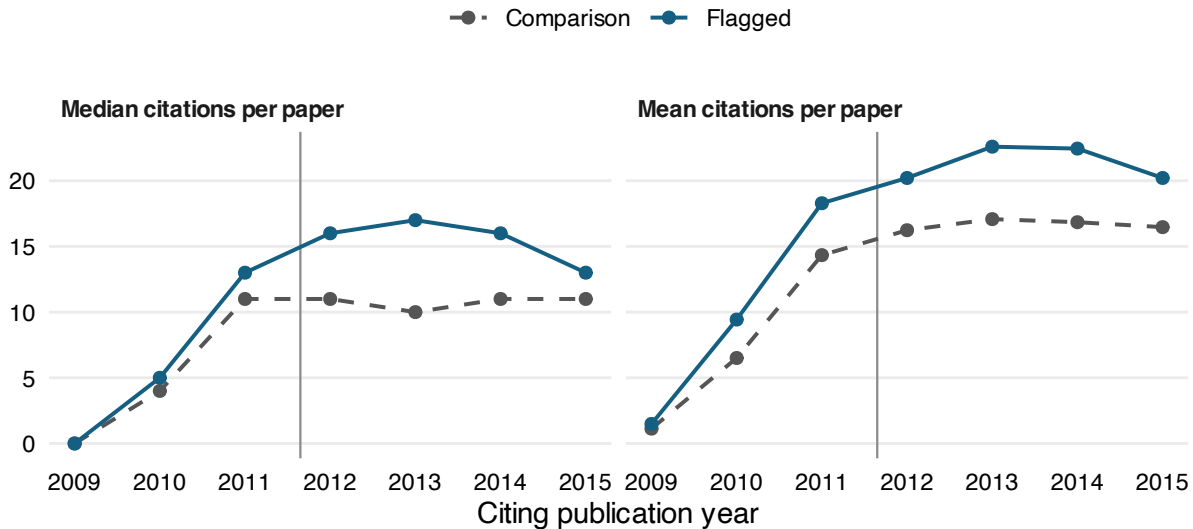


Figure 1: Median and mean annual citations to 76 flagged and 77 comparison papers, including citation-free years. The vertical line marks August 2011, when the critique appeared. The 2009 series includes prepublication zeros for papers published in 2010; the regression baseline is 2010. Lines describe the covered papers rather than a probability sample of all research.

The main regression estimates a 11.5% smaller post/pre citation ratio for flagged papers (95% interval [-31.6, 14.4]%). Thus, widespread continued citation coexists with considerable uncertainty about a relative citation penalty. The medians establish that continued use was typical, while the regression addresses how growth compares with other papers. Neither an observed increase nor an imprecise comparison establishes that publicity had no effect.

Allowing citation trajectories to differ by journal and publication cohort produces estimates from -11.4% to -7.7%. Absolute citation gains instead favor flagged papers by 1.8 citations per paper per year (95% interval [-2.1, 5.7]): flagged papers started from a higher citation level, so absolute and proportional changes answer different questions. The appendix reports the full comparisons, later response windows, and bounds on the magnitude of a relative decline.

3.2 Qualifications in citing papers

The archived citation-context sample contains 100 relationships selected after 2011, including 7 from the incomplete 2016 export. All concern citations to flagged papers. Of 95 valid completed ratings, 94 recorded no acknowledgment of concerns and 1 recorded acknowledgment. Unqualified citation was therefore common in the observed sample. The protocol records any concern about the cited paper, not only the specific interaction error.

A citation without criticism need not endorse the faulty inference. It may use another finding, a method, or background information. Nor do the codes reveal whether the authors recognized the mistake. Because the coding sample covers only the post-publicity period, it establishes the rarity of recorded qualifications after publicity, not how their frequency changed. Appendix A reports missing ratings and uncertainty.

4 Discussion

The affected papers continued to circulate after their statistical mistake was publicized. This pattern appears in median as well as mean citations and is accompanied by very little recorded qualification in the context sample. The evidence is inconsistent with wholesale abandonment of these papers. It is less informative about how many citations publicity prevented.

Citation growth already differed before the critique. Among papers published in 2009, flagged papers' increase from 2009 to 2010 exceeded that of comparison papers by 4.3 citations (95% interval [0.1, 8.6]). That short publication-year baseline cannot establish the later counterfactual trajectory. Awareness is also unobserved, and some later citations may concern findings unaffected by the mistake. These limits leave room for a citation penalty even as the affected papers remain widely used.

The practical question is how methodological criticism reaches the point of use. Readers can recognize an error from the original analysis, but doing so requires connecting a general warning to a particular paper. Our results document continued citation with little recorded acknowledgment; they do not identify which step in that process accounts for it.

References

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A Data construction and citation-context records

The supplied classification spreadsheet identifies the articles by journal, publication year, page, and study characteristics. The identified Excel and CSV versions agree on the paper-level YES/NO codes. The citation import reads 16,437 database records across the covered sheets, repairs 1 malformed record, and marks 15 duplicate DOI records within cited papers. It removes 3 citation relationships identified as false in the historical context coding. Full source hashes and record-level indicators are generated by the replication scripts.

The unreliable histories are source IDs 23 and 25. Matching the intended papers’ journal, year, and supplied page locates the target papers themselves in their supposed citing-article exports. Those histories also contain 54 records before the target publication year. This combination is consistent with exporting literature-search results instead of the intended citation results. The saved workbook contains no query log with which to reconstruct the precise operation. Deleting only the early records would not establish the validity or completeness of the remaining links. The principal estimates exclude both histories, while the appendix reports estimates retaining them. The other sheets contain neither a target-self match nor a record before the target year under these checks.

Missing sheets, source IDs 95 and 124, remain missing. In covered histories the completed 2010–2015 panel includes 29 paper-years with zero recorded citations. The scripts check that every paper has each required year and that joining the classification does not multiply or discard citation records.

Table 1: Disposition of the historical citation-context sample

Recorded status	Citation relationships
Article unavailable	1
Concern recorded	1
False citation link	3
No concern recorded	94
Uncoded	1

The sample consists of citation relationships from 2012 through the partial 2016 export. Every relationship matches a record in the source workbooks. “No concern recorded” and “Concern recorded” reproduce completed historical codes for valid citation relationships. The other categories preserve missingness and known false links.

The original coding protocol instructed the rater to assess whether a citing paper noted concerns about the cited paper, using the citing text and original article. The distributed file contains the rating and selected contextual text, not the full highlighted PDFs. We retain those ratings rather than claiming to have independently recoded the articles. The available materials contain one set of item-level ratings, so intercoder agreement cannot be independently checked.

Among valid completed ratings, the acknowledgment share is 1.1% (95% Wilson interval [0.2, 5.7]%). Assigning both unresolved, non-false records to either category gives a share between 1.0% and 3.1% among all non-false sampled relationships. These are missing-rating bounds, not sampling intervals. The Wilson interval is a binomial reference interval for sampled citation relationships; repeated source papers, selection, and coding errors are not represented by it.

B Models, sensitivity, and uncertainty

B.1 Proportional changes

For paper i in year t , the main model is

$$E[C_{it} \mid i, t, F_i] = \exp\{\alpha_i + \lambda_t + \beta F_i P_t\},$$

where F_i denotes a flagged paper and P_t denotes 2012–2015. Article effects absorb stable differences in citation levels and year effects absorb common calendar-year changes. Poisson pseudo-maximum likelihood retains zero counts and does not require equality of conditional mean and variance for the robust inference used here (Santos Silva and Tenreiro, 2006). The reported percentage is $100(e^\beta - 1)$, a relative post/pre citation change rather than a percentage change in the number of additional citations gained. In this balanced baseline specification, $e^{\hat{\beta}}$ equals the ratio of the groups’ post/pre mean citation ratios.

The covariance clusters score contributions by paper. We apply the $G/(G-1)$ and $(N-1)/(N-K)$ corrections, counting fixed effects not nested within article clusters in K , and use t_{G-1} critical values. Table 2 reports both two-sided 95% intervals and one-sided 95% lower bounds, transformed from the log scale. These are conditional model-based comparisons, not randomized-treatment intervals. No papers are excluded from the main fit for zero outcomes.

Table 2: Proportional citation comparisons

Specification	Flagged / other	Change (%)	95% interval	Lower bound
Article and year effects	76 / 77	-11.5	[-31.6, 14.4]	-28.7
Journal-by-year effects	76 / 77	-11.4	[-31.4, 14.5]	-28.5
Cohort-by-year effects	76 / 77	-7.7	[-24.4, 12.7]	-22.0
Journal/year and cohort/year effects	76 / 77	-8.2	[-24.8, 12.1]	-22.3
Retain questionable histories	78 / 77	-12.2	[-31.9, 13.2]	-29.1
Retain duplicates	76 / 77	-11.6	[-31.6, 14.3]	-28.7
Retain false links	76 / 77	-11.5	[-31.6, 14.4]	-28.7
2009 publication cohort	44 / 47	-7.2	[-25.0, 14.9]	-22.3
2010 publication cohort	32 / 30	-10.4	[-49.8, 60.2]	-44.8
Post period: 2013–2015	76 / 77	-10.7	[-31.5, 16.4]	-28.5
Post period: 2014–2015	76 / 77	-11.7	[-32.7, 16.0]	-29.7
Potentially serious errors	43 / 77	12.8	[-18.3, 55.6]	-13.9

Change and bounds are percentages, $100(e^\beta - 1)$. The last column is the one-sided 95% lower bound. Every model includes article effects. The first row includes year effects; the next three replace common year effects with journal-by-year effects, publication-cohort-by-year effects, or both. Other rows use the first row’s specification with the stated inclusion or post-period change. The baseline is always 2010. The potentially-serious subgroup retains the same comparison papers and excludes other flagged papers. Standard errors cluster by article.

The main lower bound excludes proportional reductions greater than 28.7% at the one-sided 95% confidence level under this model. This leaves substantial reductions compatible with the data. The journal/cohort-adjusted lower bounds permit reductions of 22.0–28.5%; these bounds do not resolve uncertainty about the counterfactual trend. A stratified bootstrap resampling whole papers within each group gives a two-sided percentile interval of [-31.2, 14.5]% and a one-sided bound

permitting a reduction of 28.3%. It uses 9,999 draws with seed 31415; no draws are discarded. The baseline ratio identity allows each bootstrap estimate to be calculated directly from the resampled paper-level before and after means.

Year-specific comparisons use 2010 as the baseline (Figure 2). A paper with zero citations in both the baseline and the comparison year has no finite Poisson article intercept and contributes no information to the slope; such papers are excluded from that fit and listed in the generated results. The 2011 contrast describes the transition year. The available history is too short to justify fitting article-specific trends from predominantly post-publicity observations.

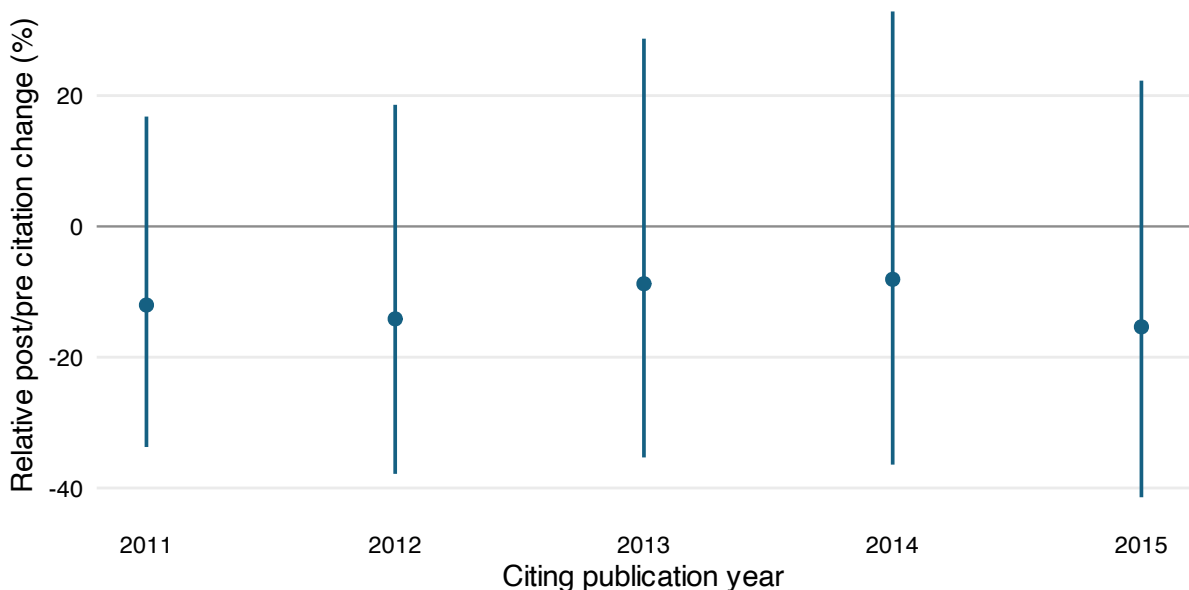


Figure 2: Year-specific proportional citation changes for flagged relative to comparison papers, using 2010 as the baseline. Points are $100(e^{\beta} - 1)$ from Poisson article/year fixed-effects models; bars are simultaneous 95% Bonferroni intervals across the five contrasts, with article-clustered uncertainty. Papers with zero counts in both compared years contribute no identifying information and are omitted from the corresponding fit. The 2011 contrast is transitional and is excluded from the main post period.

B.2 Absolute changes

For an absolute comparison, each paper’s change is its mean annual citations in 2012–2015 minus its 2010 citations. Comparing the two groups’ mean changes gives 1.8 (95% Welch interval $[-2.1, 5.7]$), allowing unequal variances of paper-level changes. OLS with article and year fixed effects on the same balanced panel reproduces this point estimate exactly; it is another implementation of the same comparison, not independent corroboration. The generated results also report its article-clustered interval. Journal and publication-year adjustment in the paper-level change regression uses HC3 uncertainty.

Removing each source paper in turn gives absolute point estimates from 1.3 to 2.5. A whole-paper stratified bootstrap gives a percentile interval of $[-2.0, 5.6]$. Medians describe typical citation

Table 3: Absolute citation changes under alternative comparisons

Comparison	Flagged / other	Difference	95% interval
Main comparison	76 / 77	1.8	[-2.1, 5.7]
Journal and publication-year adjustment	76 / 77	0.8	[-2.8, 4.4]
Retain both questionable citation histories	78 / 77	1.6	[-2.3, 5.5]
Retain duplicate database records	76 / 77	1.8	[-2.2, 5.7]
Retain known false citation links	76 / 77	1.8	[-2.1, 5.7]
Papers published in 2009	44 / 47	2.5	[-2.3, 7.4]
Papers published in 2010	32 / 30	0.3	[-6.4, 7.0]
Later window: 2013-2015	76 / 77	2.0	[-2.2, 6.2]
Later window: 2014-2015	76 / 77	1.7	[-2.5, 6.0]
Potentially serious errors only	43 / 77	4.3	[-0.1, 8.7]
Pre-critique change: 2009 cohort, 2009 to 2010	44 / 47	4.3	[0.1, 8.6]

Differences are flagged minus comparison, in citations per paper per year. Except for the final row, the baseline is 2010; the post period is 2012–2015 unless otherwise stated. Intervals are pointwise 95% Welch intervals, except the journal/year-adjusted estimate, which uses OLS with HC3 standard errors and a residual-degrees-of-freedom t critical value. The final row compares 2010 with 2009 among papers published in 2009; it is a diagnostic, not a publicity-effect estimate.

levels; neither medians nor these sensitivity checks make error status or publicity exposure random. All analytical choices are retrospective.

Of the distinct citing documents in the main papers’ 2010–2015 records, 13.7% cite more than one study paper. Shared citing documents and related topics can induce dependence across papers. Article clustering and the whole-paper bootstrap do not model those connections. The source review is also a selected set of journals and years, so these comparisons are not probability-sample estimates for all scientific publishing.